

Calculus I Lesson 07

$$1) \lim_{x \rightarrow 0} 4 = \underline{\quad ? \quad}$$

$$2) \lim_{x \rightarrow 0} \frac{2}{3} = \underline{\quad ? \quad}$$

$$3) \lim_{x \rightarrow 0} (-7) = \underline{\quad ? \quad}$$

$$4) \lim_{x \rightarrow 0} \frac{2x}{7} = \underline{\quad ? \quad}$$

$$5) \lim_{x \rightarrow 0} \frac{3x + 9}{8} = \underline{\quad ? \quad}$$

$$6) \lim_{x \rightarrow 2} \frac{2x^2 - 4x}{3x} = \underline{\quad ? \quad}$$

$$7) \lim_{x \rightarrow 3} 5x^2 \cdot \sqrt{x} = \underline{\quad ? \quad}$$

$$8) \lim_{x \rightarrow 4} \frac{2}{3}x \cdot \ln x = \underline{\quad ? \quad}$$

$$9) \text{ a) } \lim_{x \rightarrow 1} \ln(x^2 + 4) = \underline{\quad ? \quad}$$

$$\text{b) } \ln \left[\lim_{x \rightarrow 1} (x^2 + 4) \right] = \underline{\quad ? \quad}$$

9) a) $\lim_{x \rightarrow 1} \ln(x^2 + 4) = \underline{\hspace{2cm}} ?$

b) $\ln \left[\lim_{x \rightarrow 1} (x^2 + 4) \right] = \underline{\hspace{2cm}} ?$

10) a) $\lim_{x \rightarrow 1} \ln(x^{2x} + 2) = \underline{\hspace{2cm}} ?$

b) $\ln \left[\lim_{x \rightarrow 1} (x^{2x} + 2) \right] = \underline{\hspace{2cm}} ?$

11) a) $\lim_{x \rightarrow 2} \sqrt{x^2 - 1} = \underline{\hspace{2cm}} ?$

b) $\sqrt{\lim_{x \rightarrow 2} (x^2 - 1)} = \underline{\hspace{2cm}} ?$

12) a) $\lim_{x \rightarrow 3} \sqrt[3]{7x - 2} = \underline{\hspace{2cm}} ?$

b) $\sqrt[3]{\lim_{x \rightarrow 3} (7x - 2)} = \underline{\hspace{2cm}} ?$

